Active Inference LiveStream 058.0 ~ "From pixels to planning: scale-free a

Livestream | *1:46:11*

Hello and welcome. This is Active Inference live stream 58.0. October 29th, 2024. We're discussing the paper from pixels to planning scale-free active inference. So welcome to the Active Inference Institute. We're a participatory online Institute that is communicating, learning, and practicing applied active inference. You can find us at the links on the slide. This is a recorded and archived live stream. Please provide feedback so we can improve our work. All backgrounds and perspectives are welcome and will follow video etiquette for live streams. Head over to activeinference.institute to learn more about the Institute, get involved with projects like live streams, and a lot of other things. Okay, before we jump into the paper, let's say hello. I'm Daniel. I'm a researcher in California and really looking forward to this discussion. Arun, want to say hi, give any context, and thanks again for contributing to the dot zero.

Hi everyone. My name is Aaron. I'm a software engineer in the UK. I did my PhD a good few years ago in which I touched on dynamicals of modeling and active inference. I've been thinking about it for a long time since then. I came across this paper and thought it was really hard. And so I requested, guys, can we do a live stream on this? I'd really love to understand this paper properly. And now I've somehow ended up contributing to the live stream. So yeah, all good.

Okay. So we're discussing this paper from July 2024, from Pixelist to Planning, Scale-Free Active Inference, by Carl Fristen and colleagues.

The rest of this video is going to be some background and context and an introduction and an overview of the ideas.

It's not a review. It's not comprehensive. It will be part of an exciting series where in the coming weeks in the dot one and dot two, we discuss with the authors and the dot two is going to fall during the applied active symposium.

So that'll also be really fun. Okay. Let's go to the big questions. What are your big questions or whether it's about the paper specifically, or what are the big motivators that made you excited and want to pursue the these affordances? So I think the big one is, you know, the title, right? And I think the question of scale is one that's plagued a lot of active inference, implementations and previous papers.

In the past few years, everyone's like, Oh, yeah, I'd love an active inference agent that can do stuff. But if I have more than a few states, I get a combinatorial explosion.

And things are fall over or they take a very long time compared to alternative methods that are out there. So the question of scale is a really important one.

And I think this paper nominally looks like there's a way around that. Also, the renormalization group as a way to do that is, I think, a very interesting one.

It draws from a lot of physics that I actually studied a very, very long time ago. So that really spoke to me and

I thought it was really, really nice to see some things that apply to also like icing model of ferromagnetism, some other key parts of thermodynamics that now come into AI, which active inference has been doing for a while with the free energy minimization part.

But I think the renormalization group is a really interesting part of maths that seems to get you a lot further.

I also had some like much more technical questions, which maybe don't fall under the big questions, which will definitely come on to later when I was reading this paper and going, I'm looking at it and I go, I don't understand actually what's happening here.

So I'd really love to dive into those and particularly catch up with the authors in the later episodes to get some of the more technical answers on how these matrices and tensors play together, which seems to differ a little bit to the textbook by par and some of the other formulations by the RxInfer group as well.

So actually getting into the nitty gritty of how we implement this will be really fun for me.

Cool. Blue also contributed to the dot zero.

And so she wrote, what is a path more specifically, how are paths related to the renormalization group?

How is the renormalizing generative model RGM different from the active inference generative model that we usually study?

And how do paths tie into transitions between categories?

And I had similar questions.

What is the renormalization group?

How is it being used here?

How does that make the generative models structurally or functionally different?

How does that have implications for different media?

Few key claims.

They're considering one solution to the implicit scaling problem.

With scale-free generative models that are linked through the renormalization group.

They're going to explore generalized MDPs, Markov decision processes, as discrete models that do functional cognitive tasks like compression, content generation and planning.

They're going to pay attention to universality and scale and variance for specifying structural form of generative models.

And then there will be a sequence of worked examples that we're going to spend most of the dot zero walking through.

And they highlight that these kinds of RGMs with quantized paths may or may not be useful in certain application domains.

So that's kind of the meta ML question is where or how would we know where this would or wouldn't be more effective than another given kind of model?

No free lunch at all.

Could you read the abstract?

Yeah, sure.

So this paper describes a discrete state space model and accompanying methods for generative modeling.

This model generalizes partially observed Markov decision processes to include paths as latent variables, rendering suitable for active inference and learning in a dynamic setting.

Specifically, we consider deep or hierarchical forms using the renormalization group.

The ensuing renormalizing generative models can be regarded as discrete homologues of deep convolutional neural networks or continuous space models in generalized coordinates of motion.

By construction, these scale and variant models can be used to learn compositionality over space and time, furnishing models of paths or orbits.

In other words, events of increasing temporal depth and itinerancy.

This technical note illustrates the automatic discovery, learning and deployment of RGMs using a series of applications.

We start with image classification and then consider the compression and generation of movies and music.

Finally, we apply the same variational principles to the learning of Atari like video games.

Awesome.

Okay, moving on to...

Daniel, you're on mute.

Thanks.

Let's move on to the roadmap.

The first section is the introduction.

Section 2 is the background formalism.

This is going to make connections with the generative models from the textbook.

Then, section 3, 4, 5, 6 will go through a sequence of worked examples.

We're going to front load the focus in the dot zero on sections 2 and 3.

And then 4, 5, and 6 unfold kind of naturally.

Section 3 is a static image MNIST handwriting dataset task.

Section 4 is a cyclic video that then slides into a chaotic attractor dynamical system.

Section 5 and a natural video.

Section 5 goes into sound.

And 6 is the general planning case from pixels to planning.

And 7 is a short conclusion.

So, it's pretty much introduction and conclusion like bookends.

Section 2 is where the general formalism is.

And then 3 through 6 take us on from handwriting, several kinds of videos, dynamic images, sound, and then gameplay and planning.

So, here are the keywords of the paper.

Active inference, active learning, Bayesian model selection, renormalization group, compression, structure learning, RG flow, renormalizing generative model.

So, those are good terms if you're uncertain about to jump into.

Okay, we have a few slides on some of these background concepts just to get it up front.

So, we can proceed with the paper when we get there.

Let's start with what is renormalization?

So, of course, much could be said.

But, Erwin, what would you say?

What is renormalization?

Yeah, so I think this is, in itself, quite a loaded term.

There's a lot from physics.

And if you look at the Wikipedia page, the third link that's there, there's a lot in that Wikipedia page that's very confusing.

A lot of quantum electrodynamics coming in.

And I think Feynman has an attitude on it.

He says he's not a fan of it.

But I think for our purposes, the top quote that's there that's coming from the paper is actually good enough for a large part.

So, we basically take some sort of system, and we say, this is way too hard.

This is way too complex.

There's way too many moving parts.

What is a take home message?

What is the summary of what this is doing at a macroscopic level compared to the micro level?

And there are some examples in the upcoming slides, particularly talking about thermodynamics and materials, where this will become more obvious as to how this might work in practice.

But I think the coarse-grained version that retains the properties of interest is really what to focus on going forward.

And, yeah, lossy compression for the signal processing engineers in the audience.

This is very much a case of you're not keeping everything when you re-normalize.

You're keeping what's important.

Awesome.

Yeah.

As you coarse-grained so spatially, like thinking about a radius or like a convolutional diameter getting larger and larger, you're choosing a bigger chunk to grain over coarse-graining and re-normalizing what those edges and relations amongst blocks are.

So you have one model.

It's a 10 by 10 square.

So you have blocks of size one and all those edges.

And then you decide, okay, we're going to look at blocks of two by two.

You have fewer blocks, fewer edges.

So you have to re-normalize all of your parameter estimates.

And then you end up with hopefully capturing some of the signal with fewer parameters, AKA compression, whether it's lossy or lossless.

Okay.

And you, I'm going to pick up on one thing quickly where you said spatially, and you can re-normalize through space, but you can also re-normalize through time.

You can re-normalize through any dimension you want.

It's just easier to imagine the space and time ones.

So the time analogy that I think they use in the paper, which is quite a good one, is thinking about if you're, let's say you're watching TV and you're watching something.

And in that moment, there are two characters talking that might form part of a scene, which is one re-normalization.

And then you have a series of scenes, which would be an episode.

And then you'd have multiple episodes in a series.

Right?

So the time one, actually, I find a bit more intuitive to understand what re-normalization is.

And you can say, oh yeah, something happened in a particular episode.

You can say something happens in a particular series, something happens in a particular scene, but that thing you might be describing might actually be

more fine grained or more coarse grained.

Awesome.

Yeah.

The examples will start with the static handwriting and move on to include time.

And you're totally right about the timescales.

It's like saying, what's the transition matrix from minute to minute or hour to hour or day to day, but each moment is measurable with seconds, minutes, hours, or so on.

Okay.

How about the spin block transformation?

Yeah.

So now this is sort of back to material science and magnetism.

So there's a fairly well understood model of magnetism called the Ising model.

And physicists generally quite like it.

Mathematicians quite like it.

As you imagine a big magnet, there's lots of little magnets that are all line up in a grid.

And each magnet just talks to the little magnets around it.

And you can recover quite a lot of sort of actual behavior of magnets in the real world with a fairly simple model for this.

But the spin block transformation is largely going, well, if I have lots of little magnets together and they're all pointing the same way, I can just think of it as one big magnet.

That's some, that's a, some behavior of all of these little ones.

And this is a trick that we play a lot in physics of going from the micro to the macro, the micro is too hard to model.

It's too painful.

You've got all these particles.

They could all be doing, you know, spinning every which way, all their speeds and velocities.

And if you had all of them, then you could add them all up and you could figure out what's going on.

Or you can just, you know, get a thermometer and measure the temperature.

Right.

And the temperature, that one number tells you a lot about how this thing is going to behave.

So you can get a lot of value just from the macro, but that macro is composed of all of the micro.

So depending on where you are in your core screening of your scaling behavior, you can get either a lot done or a little bit done.

And it's just what's useful to you.

So I quite like this sort of quote at the bottom about the magnitude of an observable of a physical system undergoing an RG transformation.

And whether that observable is relevant or not.

I think that's quite important to bear in mind for the future examples of, well, if something is always increasing, it's always getting bigger than you're going to see it on the coarse grained transformations that you see later.

But if it's always decreasing at a particular level, you're not going to see it on the higher levels.

You're only going to see it on the lower levels.

So it will have.

So this gives you some understanding of causality of when you're going up or when you're going down.

You're going to see it on the coarse graining.

So yeah, like particle speeds and temperature is a nice example of this.

And similarly magnetism with the spin block transformation.

But by and large, what are these transforms doing?

If you take a grid of lots of little things, you just go, well, I'm just going to summarize this with a single thing that represents that behavior and then move on with my day.

Awesome.

Okay.

Quick on the singular value decomposition.

There's a nice interactive explanation.

This is related to principal component analysis, and it's a way to proportion variance and to take the biggest cut of variance for a given decomposition for a data set.

And you can take, we'll look at it in application a bit later, but it's used to assess the vectors of variance.

And then you can truncate them and use that to capture increasing amount of variance for a given data set.

And then you can take a look at it in a specific case and last kind of background topic on locality principle.

Yeah, I don't actually know a huge amount about the locality principle.

But I think it's a general intuition that things next to each other affect each other and things far away in theory don't affect you.

There are definitely some things where if this is violated, this feels very weird.

I think quantum entanglement is probably an example of that.

But I'm going to stop speaking there because we've hit the limit of my knowledge.

Yes.

I think Wittgenstein wrote something about that.

Okay.

And this is just a few of the other links, but there was a ton of other links that people brought up over the months preparing for this.

Okay.

Onto the paper.

So here's the link to the coda.

I'll put this in the YouTube live chat right now.

Just if anyone wants to follow back to the workspace where we did the preparation work.

If you're watching live, write any questions in the live chat and we'll check it at the end.

What we're about to do is go through much of the paper, pulling out some of the key quotes in the main thread.

And we show most or all the figures and some of the formalisms.

Of course, they're all there in the paper.

This is just the highlights so that we can get this done in a reasonable linear amount of time in preparation for conversations with the authors, further open source development and all that.

So, okay, here we go onto the paper.

So first three sentences of the paper.

This paper considers the use of discrete state space models as generative models for classification, compression, generation, prediction, and planning.

So for context, we're in the discrete time setting.

Figure 4.3 from the 2022 textbook.

We're in the top setting.

The inversion of such models can be read as inferring the latent causes of observable outcomes or content.

So this is the tail of two densities, meaning that generative models can be used to generate synthetic observations from latent state estimates.

That's the generative direction or decompression, as well as infer latent states from data.

That's the recognition direction from observations to latent states.

And then last of these three sentences.

When endowed with the consequences of action, they can be used for planning as inference.

So in contrast with sort of a sterile sense-making or perceptual inference task, active inference considers action as part of a joint inference distribution or problem which constitutes the agent.

So think about how the agent is defined as A, B, C, D, E, E, matrices as sense-making, latent state transition operations, and the consequences of action.

So improvements to generative modeling technologies have wide-ranging implications for a variety of both sense-making perception tasks on the inbound, as well as decision-making and action tasks on the outbound, which is kind of the theme of active inference.

So that action can be treated as an inference problem.

Any thoughts on these first lines?

No, I think we should keep going.

Cool.

Okay, specifically, this paper explores the use of generalized Markov decision processes.

This generalization in question rests on equipping a standard, partially observed Markov decision process, POMDP, with random variables called paths.

This affords an expressive model of dynamics in which transitions among states are conditioned upon paths, which themselves can have lawful transitions.

So we're going to return to this figure again.

But the big picture is that from the sort of textbook basic POMDP, we're going to add some sophistication, which is that variables called paths will condition the state transitions.

And those could fall under the agency or the efficacy of an agent or not.

There can be lawful probabilistic transitions among paths, which are like transitions amongst actions in a policy sequence, except that actions in a policy sequence would be understood to be under the control of an agent.

And those paths, states influence transitions of latent states.

Figure one shows that several concepts of depth in generative model.

We have temporal depth, which means a deeper unrolling through time.

So this is just predicting more time steps deeper into the future linearly.

So if this was each one was one day, this is just going rolling out for a bigger number of days.

Hierarchical depth would be if we had a model that had days, months and years.

And the way that it could deal with longer timescales would be have 30 days within a month and then 12 months within a year.

So nested hierarchically recursive layers deal with deeper timescales instead of needing to unroll 365 days.

And factorial depth can be thought of kind of like a multi-sensory model.

So here's an observation of sound and here's an observation of light.

And then those different modalities can be factorized separately from each other, like the what and the where in the brain to simplify.

And that's another way in which generative models can have depth.

Any more thoughts on this?

I think one thing to highlight is, first of all, the notation is quite difficult to understand in this figure.

I found it really hard.

There is also a difference, at least in my mind when I read this figure in the caption compared to say the textbook, which is the C matrix.

So I think previous formulations of actor inference, the C matrix tends to represent the prior preferences of an agent, sort of its goals, its values.

It's like, I expect to see these observations.

So if I don't, then I'm going to experience a higher free energy.

So I'm either going to change my mind or change, take some action to change the world so that my observations match my priors better.

And I get a lower free energy.

Here, the C matrix seems to be encoding probabilities of paths.

It's not entirely clear to me how we get value into that, as opposed to just what it expects to see.

So maybe there is a distinction here that I haven't understood, or maybe it is genuinely a shift.

I'd be really curious to ask the authors specifically about the C matrix and how that behaves in the future episodes.

Yeah.

Pending authors accounts.

But two things I would note to echo that first is every variable through time can be understood as a trace or a path.

Whereas path is also being used to refer to just this specific upstairs string.

And then a second case of overloading is C.

While in previous situations, like in figure 7.3 or 4.3 from the textbook, we have C as playing a role in expected free energy calculation, pragmatic value, the preference over observations.

Here, it's not the same C.

C as preferences is over observations.

Here, transition probabilities are encoded in C.

So it's kind of like a B.

But this just goes to show letters are just letters.

And future software will be able to flip out and have different shapes and notation and all of that.

So don't get too attached to the letter.

Look at how they're locally defining things, even though that can be really challenging learning too.

Okay.

So you can see a path and...

method which describes this renormalization happening across scales.

So you could have the inch by inch path and then that's nested within a foot by foot and mile by mile.

Or you could have the minute by minute path and that's nested within hour by hour and day by day.

So by having this layering of states and paths and states and paths, there's a clean way to get the separation of spatial or temporal or arbitrary scales.

So that's the big background formalism.

One other way to get an intuition.

Here's from our paper on the renormalization group in free energy principle applied to multi-scale biological systems.

On the bottom, we have the kind of figure that we've seen in many active inference papers where we're thinking about like the cognitive behavioral entity like a worm.

Where the sensory states can be associated with perception, the active states with action, internal states with cognitive states and all of that.

So at the end of the day, there's also this nested, extended evolutionary process that is happening at the population scale.

So that's kind of like nesting the phenotype of individual worms within a population of worms or within an ecological distribution.

And then last note here.

In this setting, a policy is not a sequence of actions, but a combination of paths where each hidden factor has an associated state and path.

This means there's potentially as many policies as there are combinations of paths.

So because these paths are kind of like context combinations, it means that there's a double explosiveness to them.

So understanding how do you handle that and understand the computational complexity class.

And again, where it wouldn't, wouldn't be useful.

Those are key questions.

Any thoughts?

Yeah, I think that that last bit, especially in a paper that's called talking about scale, preactive inference.

And then now we're not just combining states that explode that combinatorial explosion.

Now it's combinations of paths of states and actions.

In theory, this should be much worse.

So I think.

At the moment I'm reading this and I'm thinking we're going to pay even more of a cost in terms of computing what the hell is going on.

But I think the renormalization group and making these models hierarchical.

Pays for it.

That's that's my that's my intuition of reading this paper.

So, yeah, it's a good quote to have that.

It's a good thing to have bolded.

It's I don't think you're ever going to get a free lunch, but you have to pay for lunch.

Yeah, the number of ingredients for lunch.

Okay.

Okay.

In what follows, we illustrate applications of the renormalizing generative model RGM in several settings.

The renormalization group requires the functional form of the dynamics in terms of belief updating is conserved over levels or scales.

This is assured in variational inference in the sense the inference process itself can be cast as pursuing a path of least action, where action is a path integral of variational free energy.

Another classic term overloading action here isn't like the robot moving its arm.

This is the physics action, sometimes with a capital A.

Because all inference under free energy principle is considered in terms of a path of least action, we can be assured that arbitrarily thick, wide, thin, tall, small world, however you want to create the topology of a generative model, we can be ensured that we're in an analytical setting where the models are, quote, doing the same thing at different scales.

And then this takes us all the way from the first principles framing of the problems, all the way on through the software tools like Rx and FUR, which might be implementing these kinds of generative models also in this kind of ecumenical way.

The only thing that changes between levels are the parameters of the requisite action, e.g. sufficient statistics of various probability distributions.

That's to say, the functional properties of the RGM are simply determined and described by how it's parameterized.

So was it statistics all along? And that's what we have in Figure 3 with showing what that blocking looks like.

These are statistical parameters that are learnt.

So the description of the topology of this graph is the structure, which is either fixed or designed or set or learnt as a structure learning problem.

And then there's the values for a given situation that these variables take on.

And here's that state-path essential coupling.

That's the state-paths, state-path.

So compare this with hierarchical or multilevel Bayesian regression.

Compare this with hierarchical predictive processing architectures.

State-path, state-path.

And each of those constitute a separation of scales.

All of this is a statistical model.

Any thoughts on this?

Yeah, I think there's the hierarchy part is absolutely key here.

There is a little aside in the paper about which things in a hierarchy can affect which things in the lower part to make the inference actually possible.

So this is the idea of the Markov blankets and whether you, whether a child can have like multiple parents, that sort of thing.

I think we possibly will come onto that later.

But overall, the main critical thing is stuff happening at a higher level in your hierarchy is determining a lot of what's happening at the lower level.

But information can flow back up with the observations that you get at the lower level to infer those parameters at the higher level.

So that's the classic active inference sort of top-down bottom-up approach.

Yeah. Awesome.

Okay.

Next section, active inference, learning and selection.

This section overviews the model, which is used in the numerical studies of the following section.

And figure one provides an overview of the generative model that's considered in the paper.

We already took a quick look.

Outcomes at any time depend upon hidden states.

That's the A matrix.

That's the partially observable part.

While transitions among hidden states depend on paths.

And that's this sort of way that we see the transitions amongst latent states.

So that's the B matrix, which is intervening between hidden states S.

B is being imposed upon by this path happening.

So it's almost like this is a, this is like a, a, a rail or a guide up top.

That's influencing how the actual latent state ends up unfolding through time.

And at each time point, there's the emission of an observation.

That's the generative direction, or it could be taking in an empirical data point and then updating beliefs through time.

So, um, they describe the variables just like in the caption.

The tensors are generally parameterized as Dirichlet distributions, whose sufficient statistics are concentration parameters or Dirichlet counts,

which count the number of times a particular combination of states and outcomes has been inferred.

We focus on learning the likelihood model encoded by Dirichlet counts A.

And that's known as learning by counting.

So that's the kind of classic case of pulling balls out of an urn and then tallying every time you see a certain kind of ball coming out of an urn.

Or here it would be like pulling balls out of an urn and it's either cloudy or sunny.

And then you're doing learning by counting on the context and the outcome.

And then you're using that to parameterize everything else.

Carrying on a little bit with this and again connecting back to the 2022 textbook.

Compare with figure 7.3 POMDP.

The details are basically the same, except notice that paths U are described.

As we mentioned earlier, C in the textbook, just the letter C.

It's not actually that variable computationally, but the letter C is being used for preference distributions over observations.

Here, C is playing a role in the transition.

It's kind of like a B, but for this meta path.

In the textbook, we have a single layer model.

Now, it has compositionality, but it's just a single layer model.

Whereas in the paper we're discussing today, hierarchical levels are described in terms of this state path, state path, alternation approach.

And there's factorial depth.

Here, the procedural order of operations for the generative model is provided as policy selection, generate hidden state sequences, and generate expected outcomes.

This is the forward or generative direction.

We could also go the recognition direction from real observations to policy inference.

So they're unrolling a policy centric approach that yields expected outcomes.

That's like running the generative model from policy inference out, as opposed to running a generative model from data processing to belief updating about policy going in.

Just to highlight some kind of points of contact similarities with the textbook for those who dabble.

Any comments?

No, I think that was really good.

Awesome.

Go show.

Okay.

Equations one, two, and three in this paper are akin to equation 2.5 and 2.6 in the textbook on variational free energy and expected free energy.

So to learn about the different decompositions of variational free energy in terms of the ones that look more statistical, like divergence minus evidence, complexity minus accuracy, or the ones that look more statistical physical, like energy and entropy.

That's variational free energy.

That's variational free energy.

And then analogously with equations two and three in the paper, or equation 2.6 in the textbook, we have expected free energy with the classical decomposition into epistemic value, pragmatic value.

Okay.

Okay.

Now we get to active inference.

There's a lot we could say here.

They describe that first sentence of active inference section.

In variational inference and learning, sufficient statistics encoding posterior expectations, which could be about sense-making as well as action, posterior expectations about what I am going to do, pi, are updated to minimize variational free energy.

So that's the idea of seeing everything it is that the agent is and does on the inbound perceptual and outbound action selection as part of a joint distribution that's undergoing variational free energy minimization.

That's the active inference concept.

Note the circle with a dot notation.

So, the notation implies a sum product operator, the dot or inner product that sums over one dimension of a numeric array or tensor.

So note the difference between circle with a dot as the inner product or tensor contraction, while circle with x is the outer product.

So, to look at the figure with message passing, which brings in this circle with a dot notation.

Here, we see the same graph that we saw in figure one.

In figure one, we have a Bayes graph.

So, the nodes are variables that we would encode in our program, and then the edges are causal relationships.

So, that's the kind of classic systems modeling kind of flow interaction mind map.

Here, in this figure, we see a factor graph representation.

Now, in Livestream 55 and in the Realizing Synthetic Active Inference Agents discussion, other discussions,

we've talked more about the relationship between Bayes graph and the Forney factor graph representation.

Suffice to say that what it is that nodes and edges are basically get flipped, but there's a full reproducibility.

So that here, we have ways to look at the logistics of message passing amongst these nodes.

And it turns out that under the hood, this is what RxInfer is doing when it's being implemented.

So, maybe if we get more into implementation, we could talk about that.

This figure is just showing that for that POMDP, which is going to be doing the real work of active inference on the perceptual and action selection side,

it's not just that we can link up the variables on the Bayes graph that account for what we want to do.

It's that there's a specific logistics and we know exactly which messages to pack and send in order to get this whole variational inference done at the graph level.

And that can be done with either a determined logistical schedule or it can be done in a reactive message passing setting.

Okay. Active Learning.

Bayesian model reductions a generalization of ubiquitous procedures and statistics.

In the present setting, it reduces to something remarkably simple.

By applying Bayes rules to parent and reduce models, so a model of k parameters and of k-1 parameters,

it is straightforward to show that the change in variational free energy can be expressed in terms of posterior Dirichlet counts a, prior counts a, and the prior counts that define a reduced model.

So, A prime corresponds to the posterior one would have obtained under the reduced priors.

The alternative to Bayesian model reduction is bottom-up growth of models to accommodate new data or content.

If one considers the selection of one parent model over another, augmented model, as an action.

These two equations in the active learning section are like the add and remove variable operations and ultimately what allow us to compare any Bayesian model.

So, model comparison amongst nested, strictly nested models can be done in the fully parametric setting using the hierarchical likelihood ratio test.

Bayesian models allow model comparison amongst models that differ in only one parameter, like adding or removing one parameter as part of structure inference,

or there can be radically different model topologies underneath.

And that can be evaluated in terms of like a Bayes factor support for one model over another.

It also connects to a lot of work theoretical and applied in sleep-wake type models,

where parameters are added, fit increases during wake cycles, and then synapses are pruned, models are reduced,

core screening occurs during sleep.

And the big picture is you can use Bayesian model comparison, framed as a free energy landscape optimization just as always,

comparing different Bayes graphs, comparing different generative models,

and then this results in the expression of a log Bayes factor or an odds ratio comparing the likelihood of two different models.

So, this allows comparison of different models and the selection of different models as its own inference task.

Any thoughts on this?

Yeah, just one thing I want to pick up on is, I actually can't remember, like, I guess with this paper,

I think they were just doing sort of like, it's almost like just purely the bottom-up approach, right?

They're not starting with like a massive model and then you couldn't like start with a very big one at the top and then coarse-grained down.

You have to coarse-grained sort of going up the hierarchy as it were.

So, I think we end up going with this bottom-up growth of models because you don't want more levels than you need necessarily.

Or like, if you have an image and you're trying to classify it and it's, I don't know, 32 by 32 pixels and you just subdivide it by two,

you need to start with your pixel, you need to start with your full image width first and then coarse-grained, coarse-grained, coarse-grained up,

rather than going, ah, 8 for R, I'm going to have n levels and then cut them down after.

So, I think that's, we're probably going to be more on the bottom-up growth of models for this particular paper.

Yeah, good point. How do you know how big to make each scale and how many scales?

Hmm.

We won't read all the text on this page, but big claim.

Renormalizability is feasible under the models in Figure 1.

So, that's kind of the rhetorical chain is, here's the models that we're working with in Figure 1.

These models have renormalization properties.

That is what makes this state-path, state-path layering amenable to scale-free or scale-friendly or multi-scale generative models.

The claim is that given the renormalization properties of Bayes graphs, like or including the ones that they're describing,

with this state-path, state-path nesting, there can be computationally tractable, possibly multi-level renormalization.

And the rest of the paper, they're going to be explaining and unpacking that and adapting it to specific settings.

These renormalizing transformations have the effect such that the dynamics slow down in time and space as one ascends levels or scales.

So, compare that with other work on information theory of individuality, bottom-up and top-down information flows and causality, gapping and laddering.

And also, they're saying that the structure and the tuning of the variables in the graph, which would yield the full reproducible description of the renormalizable generative model,

itself can be reached through this Bayesian model comparison.

We had some questions about how the discrete and continuous time are different.

A more intuitive view is that successively higher levels encode sequences of sequences and implicitly composition of events or episodes.

In other words, at a deep level, one state can generate sequences of sequences of sequences, thereby destroying the Markovian properties of content generated at the lowest level.

So, this is like encoding or sedimenting nested sequences or paths in event-based or ordered structured narratives.

Deep structure is what allows for possibly advanced or naturalistic, interpretable, fill in your preferable adjective there, models of complex phenomena that do have deep structure,

like speech and writing and story.

And they're going to be walking through renormalization through space in terms of pixels, time and in general spaces.

So, it's cool to think about like Chris Fields and Mike Levin's work on competency in navigating arbitrary spaces as an invariant for analyzing cognition in diverse embodiments.

So, the idea that space in states

a higher level. So here we have a state at a higher level, s sub 1 superscript n plus 2,

and then flowing out of it, we have the state and the path at a lower level.

Higher states only change after a fixed number of lower state change, so that could be 2,

it could just be twice as fast, like a frequency doubling every time there's a time scale

separation, or it could be fixed like 60 minutes per hour, and or it could be fully reactive

in terms of using rx and fur for a just-in-time type responsive updating. And this is somewhere

where I think that generalized notation notation will be really helpful, because talking about

what these variables are in detail gets pretty messy. Because each state and path has only one

parent, the ensuring Markov blanket, the blue circles of each state, red circle, ensures

conditional independence among latent factors. There is a strict uniparental inheritance factorization

pattern in the hierarchical model, so can latent factors remain conditionally independent?

So note that the s and the u here are not connected. Conditioned upon the higher s,

they are conditionally independent. That simplifies the runtime inference. In sum,

a renormalizing generative model, RGM, is a hypergraph parameterized by two sorts of tensors,
the A and the B, in which the children of states at any level by partition into initial states and paths.
In this example, the blocking transformation groups, pairs of states and paths.
So here, it's kind of like A and B. Again, not exactly the A and the B letters from the textbook.
But note that A describes the likelihoods or empirical priors, and B is the dynamics.
So A is broadly referring to like this instantaneous sense-making-like relationship between
latent states and observations, whereas B is describing the dynamics and the transitions through time.
So I'm not sure if it's exactly that simple, but A is kind of describing synchronous
attributes of the model, whereas B describes diachronous, through-time aspects of the model.
So, okay. Anything else?
Yes. I think one thing to really hook onto is the uni-parental inheritance thing. So I think you're
correct when you say, yes, that makes everything faster and less computationally difficult. That is a
good thing. But I think there's a more philosophical thing to raise as well, which is when we're talking
about causality going up and down the levels. And learning should in theory be a lot easier
when you only have a single cause for something. So that means that when you make an observation,
it's much easier to infer, oh, the parent must have been this in order to cause what I see. And you
don't have to consider multiple factors coming in and going, oh, well, it could have been any one of 10
different factors that mean I saw what I saw. So that learning process as well, when we talked about the
directly accounts earlier, should be a lot easier to do when you have only a single parent for a particular
variable. So yeah, that conditional independence among the latent factors, I think also features into the
locality aspect as well of when we were talking about the very beginning of the session, but I'm not 100% sure on
that. But that's my intuitive understanding of what's going on there. Cool. Okay. One can also
regard RGM as an expressive way of modeling nonlinearities. In the limit of precise likelihood
mappings, this can be viewed as the composition of logical operators. One can see why an RGM can compose
operators to represent or generate content that has compositional structure. The inversion of an RGM
could be construed as a simple form of abductive, e.g. modal logic. Okay. Now we're going to get to the
examples. Here we go. First example, image compression and compositionality. So first, it's helpful for
background to look at variational autoencoders. These take in some input observations. They project down to a
latent space space that's smaller and then decode out to reconstruct. So it's really similar to a
variational autoencoder. And look at the first word, variational free energy, variational autoencoder.
They just call it the ELBO, not the VFE. There's so many similarities between this method and active
inference that it's really helpful to study up. It just sometimes uses a little bit of a frame shifted ontology.
So here's what they do. They're studying the classic MNIST handwriting data set. They map continuous
pixel values to a discrete state space, and then they're going to learn the structure through
recursive applications of a blocking transformation. They group together local pixels by tiling or

tessellating the image. So that's the first step, is they tessellate over the image at multiple scales.
So the smallest dots is a 4x4 cluster of pixels. Then the medium-sized dot is 4 of those. So then 1, 2, 3, 4, these get clustered into this one. And then 1, 2, 3, 4 get clustered into that one, and so on. So that's the spatial static renormalization via spin block locality principle process. Each group is then subject to singular value decomposition, given the training set of images to identify orthogonal spatial basis of singular vectors. So this is the grouping and the reduction. Grouping is the tessellation. It's like a convolution. We'll talk more about how it differs from convolution, like a convolutional neural net. And then they truncate. So here's an example from the form index project, but there's many other places you can look at this. This is what, for a principal component analysis or for an SVD analysis, every single time it selects the vector that explains the most orthogonal variance. And then it takes that profile out of the original data. And then it takes the next most variance explaining vector. And so you end up with something that looks like a variance explanation curve. So the first 10 components explain 70%. To get to 90% on this data set, it took 50. So if you say, well, my computer only has space for 10, just take 10. Or if you say, well, I'm interested in this kind of Bayes optimal, Pareto optimal crossover point, you can solve it that way. Or maybe you have other constraints. So that's pretty much how it's working is they're structurally using the nesting with these four by four spin glass blockings. And then they do a SVD truncation compression method on each level of that blocking.

This is how the path state, path state, path state works. It's a multi-scale blocking, convolution, tessellation, operation to group, then SVD with truncation to reduce to a defined state space. They only retain 32 of the variables from the SVD at each level.

There's more on this slide. There's one thing that'll certainly be interesting to discuss with the author is

At first glance, this procedure may appear to generate likelihood mappings of increasing size because the combinatorics of increasingly large groups could explode at higher levels. However, by training on a small number of images, one upper bounds the number of states at each level. In other words, one can effectively encode a finite number of images without any loss of information such that RGM inversion corresponds to lossless compression.

So what's the deal with training data? How do we know about what is going to be an optimal or effective data size, variability, information content, or curriculum for the data set? How do we address that data selection problem, sometimes called active data sampling?

And then to the point about the lossless compression, one can effectively encode a finite number of images without loss of information. So how can or does this do compression better than, for example, gzip or LLM? That was a great moment in computational history when people realized that in certain ways these models are doing information compression and that even gzip or just file compression utilities have attributes that trained machine learning models have because of information theory.

And how do we know and infer or describe these trade-offs related to the training data set and
the structure of the model, all these different things, as we expand the scope of our modeling degrees
of freedom? And how has this problem been solved similarly or approached differently
in other work on information geometry in variational autoencoders?
And then just last piece on working through this example. So again, they pre-processed the MNIST images.
The images were downloaded from the link here.
The image was converted. They were converted to the format that they needed.
Then they say an exemplar image is shown in figure 3 in the paper. I think that should be figure 4.
And then they're showing the centroids of each of these groups.
So this is just to get an intuition for what this multi-scale spin glass blocking is doing with the
success of clumping and grouping into these multiple spatial scales. Okay, any comments?
Yeah, I think I'm still very confused as to how this ends up being from the previous slide, how you get
lossless compression, if you're also doing the singular value decomposition and truncating some of
those vectors. I definitely want to follow up with the authors about when are we lossy and when are we
lossless? And as you said, the question of training data and how much, what goes in your curriculum,
that's going to be huge, I think.
Yeah, good point. There's a truncation of variance. So if it's close to lossless, that's also known as
lossy compression. Here is figure 5 that's describing these concatenated likelihood matrices. This is a
common method in a lot of models where four separate matrices can be included as sub-matrices in a bigger
matrix. That can make it simpler to deal with in programming. It might make it amenable to sparse matrix
methods or compression methods. It might make it just easier to visualize, but this is kind of a classic
trick going back to the SPM days. Okay. If we had imposed an additional constraint or structural prior
that likelihood mappings are conserved identically over groups. So for example, if there was the prior
that, let's just talk about these four major quadrants, that's the third level of the model.
If we had said all four of those are the same versus a situation like it was, which is it was left open,
upper left quadrant of 5 could be different than the bottom right quadrant of 5.
If we had imposed that additional constraint that the likelihood mappings are conserved identically
over groups, one would have the discrete homologue of a convolutional neural network with implicit weight
sharing. So this is a super key point because people might have a lot more familiarity with
convolutional methods from computer vision. In the case where the mappings are constrained to be the same,
so it's a situation where we're learning a single filter that optimally sweeps across a whole image,
this is the convolutional neural network case, convolutional case. Here in the multi-scale RGM,
the learning space is more complex. It's more general. So the advantage is we could possibly
describe more conditional relationships like, oh, the upper left quadrant of 5 looks like this,
but the bottom right looks like that. What's the disadvantage? You know, where is this lunch getting
paid for? It massively expands the search and training space. As always, you can have a more
expressive model, has more options, bigger search space, less expressive model, smaller search space.

And then figure 6 shows more about what these scales look like and connects it also to receptive fields in the visual processing.

But just like the 4x4 spin block window or the 32 truncation dimensions for SVD or the 10 levels of this factor, that's a big question. It's like, what about 3x3 with 25 dimensions of SVD and an 8 factor or 8 levels for the factor? So how much meta optimization do you do to determine what these variables should be?

One thing to pick up on that, which maybe they don't do in this paper, but I'm sure they've done in others is, well, if I need to optimize something about my model, why don't I just create a bunch of them and then check my free energy and just pick the one with the lowest free energy? So parameter optimization through free energy minimization happens everywhere else in active inference. I'm not sure why we couldn't also do it here for some of these.

So you'd set up your RGM, but you could have different parameters on how many variables, how many principal components to keep, how many levels to do, etc.

And so long as you can generate observations and see which ones are the most internally consistent of the probabilities based on the training data, you could probably learn which, like how to do your RGM.

Yeah. I think that's the dream is you say, here's the meta structure of the one by one, two by two, three by three, four by four, you know, n by n grouping. And then we'll learn how many n should be grouped by on how many levels per factor with how many variance retention elements from the SVD. And then you say, okay, well, should we retain the same number of SVD variables for every level? Or should we open up the possibility that, well, maybe the first level we retain 10, but the second, we only retain three. Again, that might be more expressive if there really was 10 variance contribution in the first level and three in the second, but it's a bigger search base. So that's, that's the sort of meta question, but that's cognitive computing. So it's exciting times. Figure seven, getting to the performance on the MNIST. So figure five, it says in the text illustrates the results of active learning, but I think here they mean figure seven during exposure to the first 10,000 training images of the MNIST dataset. So in this example, images were assimilated if and only if they increase the expected free energy of the RGM. So in the absence of prior preferences or constraints, this ensures minimum information loss by underwriting informative likelihood mappings at each level. So my question was, does this address the data sampling issue? It seems like they're selecting from a larger data pool, the informative examples, which improve the performance of EFE. So how much to threshold or how to prioritize that? How do you deal with a path dependence in training, the order of the curriculum of the points that are seen, or does it not matter? But for real time and online inference methods, it often really matters. It's a complex landscape. So if you head out left or head out right, it's going to make a big difference. And isn't this substituting another cost, possibly quite challenging inference and data selection problem, just so that a special smaller RGM can be made. So there's going to be situations where it is worth that entry cost to set up a smart active data sampling method to train a powerful RGM. There might be another case where the data sampling component is challenging, and it's not worth it. And there's another simpler way to do it another way. So of course, in any specific situation, it's an empirical question how these different methods perform. And then just again, with these images, figure eight.

The thing to note here is classification accuracy increases with a marginal likelihood that each image belongs to the class of numbers. So here they're showing that their accuracy is very high where they're most confident, and their accuracy drops as their confidence drops.

And from a quick perplexity search, I'll also be curious to see what others have to add. Classic shallow machine learning algorithms can easily achieve 97% accuracy on MNIST. In 2015, the main convolutional neural networks approach achieved an error rate of just 0.29%. So 99 plus percent. So especially at the way that they're presenting it here without code and things like that, keep in mind that it's more of a proof of concept in analytic foundation, not necessarily being described as a computational advance, which would have a lot more details on what was run rather than just saying it was learned automatically within a few seconds on a personal computer. That's kind of a casual way to describe it. I think it's very colorful and inspirational though. More assessment would need to be looked at.

How does the fit relate to other methods? And so this is a super exciting open source direction to support empirical reproducibility efforts to establish how and when and where different code bases do differently on different datasets.

Okay. Yeah, 100% just want to echo on the making the code and things available. We really do need that. It can't just be in a MATLAB script on Carverton's computer. It's got to be out there so that we can use it and build on it and understand what's going on.

One thing I think that's also worth highlighting is this classification and confidence question. I'm not sure if the convolutional neural networks will also give you that. So if they classify something, maybe they just say, oh, I think it's the digit zero and I'm like 95% confident. I don't even know if you get that necessarily.

Like I think maybe you get something that's a pseudo confidence number on a scale between zero and one, but it's not really confidence in the same way that this is like, this is like, you've got a probability distribution and you can do like a credible interval on it if you want. So I think that's something to really, really highlight the explainability is often in many cases worth swapping a few percentage accuracy points so that you know why your algorithm has suggested something.

Yeah. Earlier work by a lot of those authors have highlighted that true element of interpretability and uncertainty. For example, in a LLM today, even if it does say something like, I'm not sure about X, that string, I'm not sure about X was actually achieved as a high confidence outcome.

And so it doesn't actually reflect an introspection of the uncertainty intervals on X. So fun topics to go into. Okay. That was the first example. Now we're going to go faster through the next examples because they're structured much the same way. It's just each of these following examples is going to bring in a new dimension.

So the next dimension is going to be video compression. Generalize the renormalizing procedures of the previous section to include dynamics for recognizing and generating ordered sequences of images. So video is sequence of images.

Also, algorithm one has the fast structure learning. Even just copying and pasting this into perplexity and working in cursor from there does amazing things. So I'm super excited for what people come up with in terms of using the algorithms that are laid out and implementing them in different languages and so on.

So this section is going to generalize the static univariate grayscale example of MNIST into an RGB three color, three channel plus time, time color pixel voxel tuple structure.

Here's where we get to time. Time wasn't present in the MNIST example, but by partitioning each sequence into non-overlapping pairs during the timescale transformation at successive levels,

one effectively halves the length of the sequence when ascending from one level to the next.

So we talked about different ways to do nesting in time, 60 minutes in an hour, 24 hours in a day.

Here is going to be a strict frequency doubling.

This means that higher levels encode sequences of sequences of sequences,

paging exhibit, that can be regarded as episodes of successive events.

So key idea.

State path, state path, state path, state path encodes transitions through time,

and there's this enforced frequency doubling.

So processes are encoded relationally.

Look into something like successor representation learning or cue learning to see some other approaches.

So, it's possible to define and use message passing among levels.

It could also be possible to provide alternative operations orders or, like in RxInfer,

use graphs that have no preset order of updating or other rules for how updating occurs.

Okay, here's where in figure 10, 11, and 12, we get to the example.

We used a short video sequence of a dove flapping her wings.

The original video was downsampled to 32 frames, and they describe a little bit more.

They used the same 4x4 blocking.

So, this is key information on the video processing.

And it's really exciting to think about what they do to these file formats as analytic affordances.

I found this actually more from the discourse analysis area and rhetoric studies.

But if we think about the choices and the degrees of freedom that are made in the data processing and meta-modeling,

those are analytical affordances.

And it relates to how we were talking about you could do inference on those variables.

And so, here we see analogous to the images that were output for the MNIST summaries and performance.

Here, we similarly have prediction through time of this dove flapping the wings.

It's a cyclic generated video.

And, for example, the stimulus here is just if a spotlight is only on part of the dove.

And then it can reconstruct through space at the same moment and through time

how it will continue to unfold given the patchy observations that it gets.

So, that's the kind of dove video example.

I think there's one thing just to, yeah, some background on why does that matter?

That you can just all do the spotlight thing and recover the original in a way.

So, this is a fairly common way of understanding sort of like unsupervised learning.

Has your algorithm actually understood the data?

And so, masking parts of the image and asking it to infill it is a very common way of testing
that what you've got actually understands what's going on
and it's not just going to hallucinate anything random.
So, yeah, when I first came across it, I was like, oh, this is a bit odd.
But then, yeah, it's worth looking into just as background of like,
well, if you have an unsupervised learning technique,
how do you know it's actually doing a good job?
Yeah. Cool.
Okay, the next example in the video section.
In the preceding example, the sequence of the image constitutes a closed path,
namely a simple orbit.
So, it's like a gif of the dove flapping the wings.
It returns to the same spot and then it continues again.
This can be regarded as a quantized representation of a periodic attractor.
In the section paths, orbits, and attractors,
they use the same procedures to compress and generate stochastic chaos
using images generated by a Lorenz system.
The aim of the example is to show how active learning after structure learning
furnishes a generative model of chaotic orbits.
So, Lorenz system and active inference is discussed in the four-part series on Livestream 32,
stochastic chaos and Markov blankets.
And here we see an analogous process to the MNIST, analogous to the dove.
We have an image or sequences of images through time.
It's traces from this Lorenz chaotic attractor.
Similarly, there's this grouping and reduction operation happening through space and time.
And then, parameterization of those models, that's the structure learning followed by parameterization,
results in the ability to have, even when there's a gap in the observations,
like in figure 15 on the right,
here we have a gap in the observations, but the predictive posterior is able to continue.
It's an empirical question how well it lines up,
but this is just really fascinating because it shows, well,
even though the generative process, we can say,
has this chaotic instability
that might make it analytically intractable,
from the generative modeling perspective,
the RGM enables a really nicely typed way
that can accommodate systems that have these chaotic properties.
What was exciting to you about this part?

Yeah, exactly that.
It's, you know, chaos is fascinating.
And modeling it, famously very difficult.
So, yeah, I think this is actually,
I feel like they should be shouting about this more in the paper, in a way.
Like, maybe they could have opened with this.
I don't know.
But I would really like to see some more,
like actually simpler examples
of chaotic systems being modeled in this way.
And maybe that could be a separate paper entirely.
But even just, you know, one-dimensional signals,
that change in time,
that are famously quite chaotic,
that we don't do a very good job of modeling,
would be really interesting to see how getting that into an orbit modeling procedure
using this would be and see how that goes.
We were very excited to try that.
Awesome.
Yeah, totally agree.
The next example,
continuing in the image and video sequence,
is a bird.
So, a natural extension.
Figure 16 illustrates the spatial blocking of a more natural video,
a short sequence from this website.
So, that URL was no longer active.
But either way, it was just a search URL.
So, it'd be cool to see the exact video.
Maybe it could be uploaded.
But either way, this is a short video
of a bird in a natural setting.
And again, figure 17,
staying with this theme,
there's a latent state prediction,
ongoing,
even though the stimulus,
the observation,

is withheld sometimes.
So, in a way,
that's like what happens when we blink
or have an eye saccade.
We have this ongoing posterior prediction
of the visual scene,
which is what's being experienced or attended to,
with color everywhere,
and equal precision everywhere,
and all this stuff,
including continuity
when we blink or have an eye saccade.
Because it's basically what's happening here.
There's just transiently no stimulus,
but still the likeliest thing that's happening
is the latent state prediction.
So, that last little sequence
took us from handwriting,
static images, grayscale,
into three video settings,
which was a cyclic,
oscillatory, synthetic video,
a synthetic,
chaotic video,
and then a natural video.
So, you could imagine if this was
more structured natural video,
it would be easier to learn
with probably any method.
If this was a more complex scene
with multiple different things,
switching positions,
it would be a harder video to learn.
Any more thoughts on the natural extension?
Yeah, actually,
so, if it's trained on this particular video,
one thing I wasn't sure is
what frames does it predict

sort of at the end?
So, we've talked about like,
oh, if you like,
miss some frames in the middle,
and then it can catch up and keep up.
But if you just run this indefinitely,
I'm not actually sure
what the bird would be predicted to do,
and would it just eventually loop back?
Because it's sort of like,
if you had that chaotic path
in the previous one,
it's still in orbit.
It still returns to the same attractor states.
Whereas, here,
I'm not sure what the bird would do
when it's predicted to be doing stuff.
Maybe it just flies off the screen
and we don't see it anymore.
I don't know.
But that's,
it's kind of like a sci-fi.
It's a simulation to run.
It's like a sci-fi.
Like, what if,
it's like the movie's an hour and a half
long,
and then you keep watching,
it's like,
wait, it's still going.
What's happening?
Okay.
The next case that they study
is sound.
So,
the video examples
in the prior three
were just transitions.

It was an ordered sequence of images.
Whereas,
the video,
more broadly understood,
like the live stream
that you're listening to now,
is video as image plus audio.
So, here,
they just kind of
leave the video behind
and say,
let's just look at the audio.
In this setting,
pixels,
and there's an analogy
for the multimodal LLMs,
where the tokenization methods
that work for natural language
in the discrete setting
also work for tokenization
of image
and audio
like GPT-4O.
So,
in this setting,
pixels,
i.e. picture elements,
are replaced by voxels,
volume elements
over frequency and time.
These constitute
time frequency representation
of time series
or a continuous wavelet transform,
CWT.
Renormalizing generative models
for sound
are simpler

than for video content
because there's only
one metric dimension,
frequency,
that accompanies time.
using a spin block transformation
to coarse grain
over frequency
reflects the factor assumption
that neighboring frequencies
fluctuate in a correlated fashion.
So,
kind of similar
data availability setting,
URL provided
but unclear
what the songs were,
and then we see
the similar
visual representations
culminating in,
for example,
20,
where predictions,
latent state predictions,
are able to continue
even though
part of the stimulus
has been redacted.
So,
it's kind of like
a video chat
or a phone call
and then someone
drops out
for a few seconds,
but you're able
to still have

your unrolling
expectations
about what did happen
even though
you might be missing
some observations.
So,
any comments
on
this first sound part?
No,
let's carry on.
All right.
All good.
From sounds to music.
One might ask
if RGMs
could be applied
to language,
perhaps in the spirit
of hierarchical
Dirichlet process models.
We will not
address this here,
but provide
an application
to music
to illustrate
the generalized
synchrony
that accompanies
communicative exchange
Fristen and Frith 2015,
the classic
songbird dyad.
So,
keep in mind
that the birds

that are in this
dyadic
songbird setting,
which is also
reflected in the
2022 textbook,
are literally
singing from
the same sheet music.
They have the same
single fixed song.
So,
it's not
modeling
the flow
of language
as conversation
at the level
of new
lexemic
information provided,
like go to the store
and get me
milk,
or
meta
lexemic
like prosodic
or stylistic
elements
of language.
It's a very
restricted
sense of language,
but it's part
of the tech
tree of
building towards

models that
capture more
sophisticated
elements of
language,
but here
is more
structured
than this
sound example.
It'd be
interesting to
hear some
of these
songs in
their original
versions and
in their
generated
versions,
and like,
how well
does this
do
from a
performance
and file
size
perspective,
how well
does that
do to
MP3?
Which is
another
example of
data compression,
and that's

exactly like
the GZIP
and LLM
comparison
earlier.
if at the
heart of a
variational
autoencoder
or an
RGM or
something else
is using a
structurally
determined
model as
parameterized to
have a
smaller
representation
of a
larger
space,
how do
these
physics-inspired
first principles
of intelligence
models or
classical
machine learning
and compression
related models
do,
are we
getting
what fraction
of compression,

how good is
the performance,
all these,
you know,
can you tune
the bit rate,
all those
kinds of
questions.
Okay,
on to the
last
section
from pixels
to planning.
So,
in the
final section
we turn to
the deployment
of RGMs
for planning
as inference
and implicitly
their use
as synthetic
agents and
decision makers
under uncertainty.
So,
we're getting
to the
generalized
active
agents,
which is
to say
the capacity

for control
and navigation
in arbitrary
spaces.
And so,
here we
also see
a comparison
or juxtaposition
of active
inference
and reward
learning
and ran
way,
give a
great
guest stream
recently
on similarities
and differences
there.
And they
make some
more
comments
about
how
the
RGM
has
relationships
with
generalized
multi-level
learning.
So,
now we're

kind of
briefly
dropping
any specific
case,
like a
data set
of images
or video,
and just
thinking
more
generally
about
how
this
RGM
could be
used
for
multi-scale
active
agents.
Any
comments?
No,
it's
carry on.
Procedurally,
this looks
very much
like
conventional
reinforcement
learning.
However,
it is
simpler and

more efficient
because it
learns a
compressed
representation
of and
only of
paths
that link
rewarded
states.
So,
that might
have some
connections
with the
inductive
behavior
paper that
they also
wrote.
Another way
of looking
at this
is smart
data
selection.
That relates
to my
earlier
question of,
well,
if you knew
which data
to select
and in
what order
for free,

then maybe
the model
can be
trained
effectively.
But,
if it's
a hard
problem
to determine
which data
to sample
or what
order to
do it
in,
then it's
just
substituting
one type
of problem
for another,
which might
be effective
in a situation,
but that's
not a general
approach.
Effectively,
this leads
to the
automatic
selection
of model
structures
that can
only recognize
and predict

rewarding
episodes,
thereby
eluding
subsequent
parameter
learning.
This rests
upon the
fact that
one does
not need
to learn
how to
maximize
reward
if one
has a
generative
model
that can
only
predict
paths
that lead
to reward.
I was a
bit uncertain
about this
because
different
paths
may
probabilistically
lead to
different
reward,
and then I

thought,
well,
wait a minute,
that's just
pragmatic
value.
What about
the automatic
selection of
model structures
that can
only recognize
and predict
epistemic
information?
isn't the
expected
free
energy
going to
be able
to select
data,
active
data
sampling,
live stream
57,
smart
data
selection,
according
to epistemic
plus
pragmatic
value,
and thereby
specify the

next action?
Again,
reinforcement
learning selects
the next
action based
upon highest
expected
value or
reward,
but expected
free energy
has epistemic
plus pragmatic
value.
In what
follows,
we will
illustrate this
approach in
sequential policy
optimization and
unpack some
of its
corollaries.
Games and
attractors.
So here,
they use simple
Atari-like games
to show how
a model of
expert play
self-assembles
given a
sequence of
outcomes under
random actions.

The first
simple game in
question is the
game of
Pong.
They describe
how they
core screen
the blocks,
and then
they selected
frames that
started from
a successful
outcome.
So they
trained on
successful
examples.
In short,
we use
rewards for
and only for
data selections.
That's kind
of an
interesting
possibility
that there's
a pragmatic
filter on
training data
followed by
then an
epistemic
openness to
whatever it
is that that

data has.
renormalization
happened through
space and
time like the
video examples
and through
the action
policy.
That's the
section that
we're in here
where we're
saying in the
general case,
we're also able
to do RGM on
nested policy,
but here there
was just one
timescale of
policy.
figure 25,
26,
and 27 go
over some
of those
representations.
So any
thoughts on
this?
Yeah, I
think it's
the training
part is
really
interesting.
and what

I think
maybe possibly
as well,
we're getting
a bit close
as to why
they use
that C
letter for
paths.
If it's
only ever
encoding
paths that
are rewarding,
then we
sort of go
full circle
back to the
definition in
the path
textbook of
C is like
rewards.
It's just
rewarded paths
rather than
rewards of
observations.
But I think
they come onto
this later on
in the paper.
This feels
like a very
brittle way
of training
a model.

If it only
ever encounters
positive states
or things
that it likes
or things
that we
want it to
like,
what happens
if you drop
it in a
situation that
it hasn't
seen before
where there's
a negative
state and
it has no
way to
recover back
to a
positive one?
So if you
think of
for training
on a game,
if you start
in a losing
state,
how do you
bring it
back?
How do you
recover?
So I think
that's something
that we will

come on to
later,
but it's a
really interesting
thing to bear
in mind of,
well,
what are you
training on?
Are you only
training on
success?
How do you
deal with
failure?
Yeah,
great point.
That also
makes me wonder
about the
transfer
applicability.
So we train
it on that
bird video,
but then what
if it's the
same bird,
but it's just
a little bit
darker light,
or it's a
different angle?
what are the
transfer limits,
or what is it
really training
on?

And then
especially if
there's a sort
of data
filtering or
selection to
pass data,
that is one
of the
critiques of
large data
set related
training.
So maybe
this uses
a subset
of it,
like a
filtered
subset,
since there's
already a
database of
successful and
unsuccessful
plays,
just selecting
the successful
ones.
And then at
the same time,
there's this
interesting
biological tie-in,
which is like
the organism's
training data
through life,

in a way,
is all
successful with
respect to
survival.
So by
having the
training data
only reflect
a certain
kind of
success,
perhaps with
a lot of
variability still,
it's an
interesting question
what it
learns and
how
transferable
that could
be.
And then
this is the
last of the
worked examples.
Figure 28
shows the
same simulation
using a
slightly more
complicated game
based upon
breakout.
So what's
happening in
these gameplay

settings
is gameplay
is like
a video
plus
controllable
state
transitions.
So it'd be
like if there
was a dove
flapping its
wings,
and then
also there
was another
parameter,
which was
lights on
or lights
off.
And then
there's
one latent
state,
which is
where is
the dove's
wings
really,
and then
how does
the dove's
wing state
get emitted
to observations
when the
light is

on or
off.
And then
that would
be like a
simple example
where the
dove does
not respond
to the
light.
And then
a more
complex
example would
be,
oh,
it flaps
differently
in the
light and
the dark.
And so
that's where
you start
getting into
these
factorized
models and
paths on
states on
paths and
states.
So here
they study
this game
Breakout,
where there's

in a way
niche
modification
happening.
it doesn't
show any
of the
cells on
the top
getting
broken,
but that
would be
interesting.
How does
it represent
down into
the really
fine scale,
what numbers
represent the
presence or
absence of
a block?
And how
does that
really get
encoded?
So,
any last
comments before
we go to
the conclusion?
I actually
found this
example the
most difficult
to follow in

the paper.
I think
it's
still not
entirely clear
to me how
this RG
operator
handles the
action inside
this as
well,
and how
you
coarse-grain
the effects
of action
that might
happen at a
particular
time step
that then
gets sort
of diluted
out,
as it were.
So,
that's just
something I
think for
the game
section that
I'd quite like
to ask the
authors about
is to go
into a bit
more detail

on the RG
operators
when action
is happening,
but that action
can only happen
on a particular
time,
like,
what's the
word,
sort of
period,
as it were.
Yeah,
also,
I wasn't sure
if these were
meant to be
empty or
something,
but then I
wondered,
well,
is there a
natural
interpretation
separation of
levels 1,
2,
and 3,
and do
these scale
separations
capture some
natural
components of
the game,

or is
there a
possibility that
by using
like a 4x4,
any nxn,
or using a
strict doubling
of time,
that there's
kind of an
aliasing where
real natural
scales of
analysis
get sort
of chopped
up strangely
with different
timescale
separations,
and so
then it
depends a
lot on
what
spatial and
temporal
scales are
divided on,
and then there
would be some
training data
for some
partitionings
that would
work really
well,

like if you
had a ton
of good
information on
the structure
of a
problem,
then maybe
one measurement
would be all
you needed.
Like if you
knew the
whole class
schedule of
the university
and how all
the class
beginnings and
endings were
related,
and then someone
said,
there's a lecture
in that hall
on Tuesday,
and then you
could have a
ton of information
about how that
was correlated
with everything
else.
But then if you
were studying it
on a meter
by meter
subdivision

on a timescale
that wasn't
related to the
timescale of the
classes were
happening,
you might
need a ton
of data
and still
be shocked
with the
next time
step.
So that's
just the
structure
learning
challenge,
and I
think that
is the
hope and
the question.
How can
automated,
semi-automated,
and ultimately
unified methods
for structure
learning be
applied?
Okay.
Conclusion.
The paper
has showcased
several applications
of a generative

model for
discrete state
space approaches
based on the
renormalization
group.
The appeal
to the
apparatus of
the
renormalization
group is
relatively
straightforward
in the
setting.
This is
because
active
inference
is an
application
of the
free
energy
principle
which is
itself
a
variational
principle
of least
action
whose
functional
form is
conserved
over scales

or hierarchical
levels.
Having said
this,
the restriction
to spin
block
transformations
and pixel
spaces
could also
be regarded
as a
limitation.
Classic
answer.
In
principle,
any
outcome
space
could be
a
candidate
for
course
graining.
How do
we know,
understand,
and balance
where and
when these
spin blocks
would work
or not?
We've
discussed

in other
conversations
the applicability
of the
viability
of doing
sweeps
and structure
learning
and
act
inf,
on act
inf,
on act
inf,
on RL,
on act
inf,
on RL
and how
many layers
back
before your
data science
team is
just like,
I think
this is
fine.
RGM
with quantized
paths may
or may not
be useful
in certain
application
domains.

The
applications
above
illustrate
the simplicity
and efficiency
with which
the structure
of the
models can
be learned.
So that's
the key
claim.
That's
the key
point.
Further
limitations
of the
approach
described
above rest
on the
efficiency
of the
accompanying
methods.
The
efficiency
is both
a gift
and a
burden.
Again,
great.
Clearly
in terms

of sample
efficiency
and compression,
the schemes
described
above are
designed
to outperform
conventional
learning
schemes.
Okay,
but designed
to outperform
is something
that anyone
can say.
So,
did they
outperform
when other
learning schemes
weren't presented
in the paper?
This follows
because model
construction
procedure is
constructive.
as opposed
to the
reductive
approach
of Bayesian
model
reduction
and related
procedures.

Enabling
the self-assembly
of a minimally
expressive
model to
recognize,
generate,
and predict
the content
at hand.
So,
that's this
bottom-up
construction
through
tessellation
that we've
discussed.
This should
be contrasted
with the
reductive,
but not to
be confused
necessarily
with reductionism
or reductionist
account,
but reductive
in terms of
starting with
a large,
overly expressive
model that's
reduced or
pruned,
minimizing
complexity to

recover predictive
validity
and generalization.
So,
one way to
think about
that is
like in
the
VAE,
the latent
space is a
bowtie.
It's smaller
in the middle.
Whereas,
for an image
processing neural
network,
you might take
in 64 pixels,
output 64 pixels,
but then the
thickness in the
middle is huge.
So,
these are the
approaches saying,
well,
you can kind of
have this
constructivist
approach to
building a
smaller latent
space that's
projected down
to and then

unpacked from.
That's the
VAE,
RGM type
style.
Another
approach is to
start with a
massive surfeit
of parameters,
you know,
70 billion,
405 billion,
and then use
that as a
function approximator
to fit
an approximation
to some
other function.
But,
more generally,
how could
efficiency be
a burden?
They make
some more
points about
learning
and skills,
and this gets
to our
question about
the successful
trials for
training.
So,
these such

models are
necessarily
brittle
in the sense
they will not
recognize or
respond to
events they
have not
previously
encountered.
That almost
seems like
too far to
say.
I mean,
the MNIST
model will
be able to
respond to
a digit it
has never
encountered.
It won't be
able to
respond to
a temperature
reading because
it just doesn't
even have that
capacity, but
it's like out
of sample
learning or
out of
structure
learning.
A useful

heuristic here
is learning to
ride a bike.
In this
kind of
procedural
learning,
we learn
from a
starting
position and
accumulate
successive
cycles of
pedaling until
we avoid
falling
completely,
but we
do fall
and we
learn from
when we
fall.
In other
words,
we retain
only experiences
characteristic of
successful bike
riding until
we can ride
fluently.
However,
we would
not be able
to ride a
bike starting

from any
arbitrary
state,
e.g.
falling
over.
We would
have to
return to
a known
starting
position to
recover our
flow.
Flow is
meant in
manifold,
which itself is
a pun,
senses here,
the fluency
associated with
being in the
flow,
control flow,
dynamical flow,
and of course
RG flow.
So what is
really being
claimed about
what are the
situations where
successful examples
are enough,
where negative
examples are and
aren't learned

from or
recovered to,
and then how
well do these
methods work
within and
out of sample?
After all,
that is what
the whole question
is in training
a model,
is how well
do you do on
the training
and the test
for different
kinds of
out of sample
being from
really similar
but not in
the exact
training data
itself,
all the way
on through
totally different
kinds of
work.
So,
we don't know
the next steps
or implications,
we have a lot
of questions
and hopefully
do too.

What are you
looking forward
to or just
what are your
thoughts having
now gone
through all
this?
I'd love to
see some
code.
I'd love to
see some
implementations
that we can
hack around
with ourselves.
and yeah I think it's a really because a lot of what's come in the past for active inference has
been it's felt very like I need to write my generative model down on paper I need to really
sit down and think undistracted and think how do I model this problem what are the causes what are
the observations how do they how might they relate and then I create a function which will generate
these observations and then you can plug it into active inference and go all the way but the often
the hardest part is writing them writing down those a b c d matrices right it's actually really difficult
from lots of people you know you can be a trained physicist and it's still really hard to think how
do I model this problem in the real world in terms of mathematical matrices that's really hard so in
theory if your algae operators are good enough and you just you can just chuck some data maybe it has
to all be positive data all the happy path stuff but that's already quite powerful I think and moves
I think active inference closer to more traditional machine learning in terms of accessibility and ease
of use so that data science teams could just use this in production I think what this is this paper
does represent quite a significant step forward for that and I think you know with all the work that
you've done on this channel of just making active inference more accessible and easier to play around
with easier to try in new scenarios that's that's such a good thing so very excited for the future
live stream sessions on this paper awesome yeah great point that how much of the modeling specification
today and tomorrow will be manual semi-automated automated like would you just put in the EEG data
and just say go for it or how would you embody side information about the EEG problem and then how would it do on
training and test and how would it transfer all these different questions so it's great do you have any last
comments uh yeah I just think the explainability thing is going to be the key here so as soon as you

go to unsupervised sort of structure learning what did those levels mean and what did those factors mean that it's learned from the data itself if I if we go the old-fashioned route if I come up with a generative model and I say this is how I'm modeling the system I can explicitly say this level this latent variable is for whether it's raining or not and this observation corresponds to water observed on the graph outside but because I'm coming up that model I can use words to communicate that to you and you have that full explainability of this model with this approach we are going to lose some of that and it'll be really interesting to see how whether we can recover the explainability of well what does this factor mean when I did a principal components transformation and ditched half the components what does that actually mean you can sort of look at it on the MNIST example and go oh yeah it's a digit right but we still have to look at that so getting that in a semi-automated way would be very exciting as well yeah that makes me think about what natural joints of the world handles for the world will a possibly automated method fail to find and then conversely what new natural scales might it come up with for example just doing PCA analysis on natural language it's like okay principal component one is whether it's happy or sad or whether it's written in this language or that language but then you get to principal component seven and it's like huh does that eigen term mean something with it a little bit more of this and a little bit less of that does that what does that really mean and then the further you go out usually the less reliable those variance components are across different data sets so then that comes back to how do we interpret and set up systems that have few or many you know when shorter lists are more interpretable but then when you clump under fewer things they have more grouping of dissimilar things so it's like lumpers and splitters the neats and the scruffies those those are all coming to play right on that field so thank you very much arun and also to blue and to courtney for the contributions to the dot zero so looking forward to the next two weeks peace thanks very much
thank you